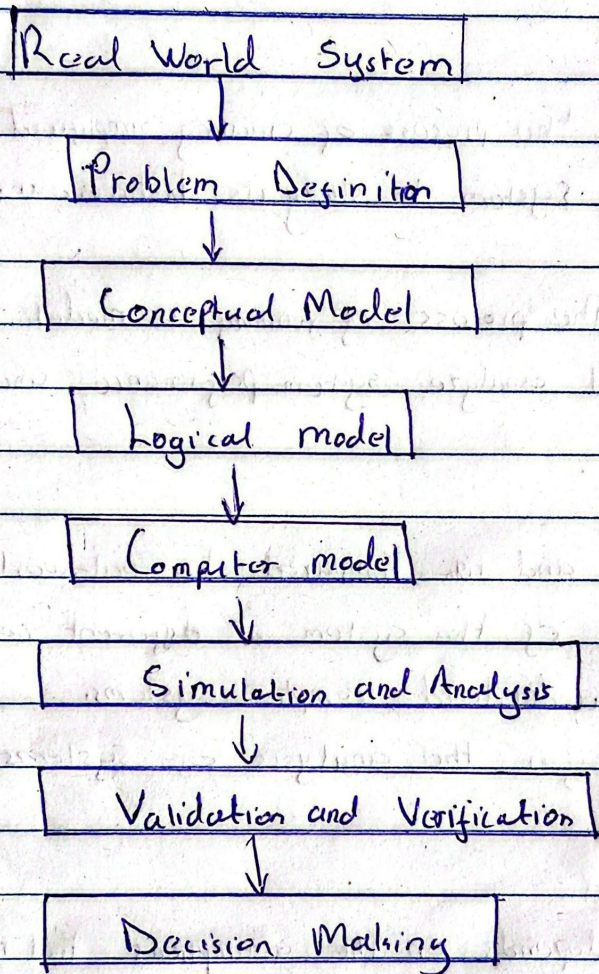


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CSC 31H

a) modelling process Diagram.



Explanation:

Problem definition: here you find the real issue and the objectives of the model.

Conceptual model: Here a representation of the model is made using diagrams, assumptions or relationships.

Logical model: here the concepts are translated to equations, rules or algorithms.

Computer model: The model is implemented using softwares or programming tools.

Simulation and analysis: tests are made to observe the system behaviour.

under different conditions.

Validation and Verification: here the model is confirmed to ensure it is accurate and behaves as expected.

Decision Making: The results gathered all through are used to support the planning and making of decisions.

b) Modelling is the process of creating simplified representation of a real world system to study its behaviour.

Simulation: The process of running a model over time to observe and analyze system performance under various scenarios.

Advantages:

- Reduces cost and risk compared to real-world experimentation.
- Allows testing of the system at different conditions.
- It helps to understand complex systems.
- Saves time for the analysis of systems.

Disadvantages:

- The results depends on the assumptions that may be bias.
- It requires skilled personnel which is expensive.
- Does not necessarily represent reality.

c) Application areas of modelling and Simulation.

- i) Engineering and Manufacturing: used to test the machine performance.
- (ii) Healthcare: used to model the patient flow and to study disease spread prediction.
- (iii) Business and Economics: used in forecasting the demand and sales.

d) Guidance on validating the first-time model:

- Compare model results with real-world of historical data.
- Perform Sensitivity analysis.
- Review the model with domain experts.

e) Advantages and Disadvantages of Monte Carlo Simulation

Advantages:

- Flexible and easy to implement
- Works well with complex systems.
- Useful in risk analysis
- Handles uncertainty and randomness effectively

Disadvantages:

- Require many simulations which expensive.
- Results are approximate, not accurate.
- Depends heavily on input probability distributions which is complex
- Can be time consuming.